

BodySys 2025: The 11th Workshop on Body-Centric Computing Systems

Qiang Yang
University of Cambridge
Cambridge, United Kingdom
qiang.yang@cl.cam.ac.uk

Thivya Kandappu
Singapore Management University
Singapore, Singapore
thivyak@smu.edu.sg

Abstract

The 11th Workshop on Body-Centric Computing Systems (BodySys 2025) aims to advance the discussion and innovation around wearable and body-centric technologies for mobile and ubiquitous computing. Building upon a decade of successful workshops, BodySys fosters interdisciplinary dialogue among researchers and practitioners across sensing, communication, AI, health, and human-in-the-loop systems. The 2025 edition will feature keynote talks and ten technical presentations, addressing emerging topics such as implantables, intra-body communication, smart fabrics, and AI-powered wearable computing. The hybrid format encourages broad global participation and positions BodySys as a catalyst for shaping the future of human-centered mobile systems.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools.**

Keywords

Mobile Computing; Ubiquitous Computing; Body-Centric Computing; Pervasive Computing

ACM Reference Format:

Qiang Yang and Thivya Kandappu. 2025. BodySys 2025: The 11th Workshop on Body-Centric Computing Systems. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3714394.3750559>

1 Introduction

The BodySys workshop centers on the latest developments and discussions in wearable and body-centric computing technologies, spanning both human and animal applications, and their impact on mobile systems and applications research. Initially established as the Workshop on Wearable Systems and Applications (WearSys) and held under that name for six years, the workshop expanded its scope in 2021 to encompass the broader domain of body-centric computing. This transition positioned wearable computing as one facet of a more inclusive agenda that also embraces systems for animals, such as implantable devices in pets. BodySys aims to create a vibrant forum that brings together researchers and design

practitioners to explore how wearable, body-integrated, and user-in-the-loop technologies can enrich mobile systems research. It also serves as a launchpad for bold, forward-looking ideas in this rapidly evolving field.

The BodySys workshop arrives at a pivotal moment, as body-centric computing, encompassing sensing, communication, and computing systems, has gained increasing prominence within the mobile computing landscape. The widespread availability of commercial wearable devices, emerging capabilities such as 3D-printed biological components (e.g., tissues), and the integration of human users into AI-driven decision-making processes are collectively reshaping the direction of mobile and wireless systems research. This vibrant and interdisciplinary domain intersects with areas such as human-computer interaction (HCI), wireless networks, healthcare, fashion, and sustainable energy, and is now driving technological innovation at the convergence of these fields. With rapid progress in wearables, implantable technologies, wireless communication, and human-in-the-loop systems, body-centric computing is becoming more relevant than ever. Furthermore, the adoption of smart sensing and diagnostic tools is expanding into domains like animal health and smart agriculture.

BodySys aims to serve as a catalyst for continued innovation in this dynamic space, offering a platform to showcase bold, forward-thinking ideas and to help steer the future trajectory of research in body-centric computing systems.

2 Workshop History

The current proposal is towards organizing the fifth iteration of the BodySys workshop which we envision as an expansion of the WearSys workshop (2015-2020). This version marks the 11th year of the efforts from this steering and organizing committee to present a workshop on an emerging research area, and it looks forward to continuing its tradition of co-locating with UbiComp 2025.

Since its founding in 2015, the workshop (initially called WearSys, now BodySys) has evolved significantly over the past decade. It has been organized by leading researchers from institutions such as CMU, Microsoft Research, TU Delft, Nokia Bell Labs, and others. The workshop expanded its scope over the years, from wearables and mobile sensing to broader topics like smart fabrics and body-centric computing, leading to its renaming in 2021 as BodySys. It has maintained strong participation, with consistent paper submissions, a steady number of invited talks, and an engaged audience. The workshop is also known for featuring distinguished keynote speakers each year, including researchers from top universities (e.g., CMU, UNSW, ASU, NUS) and industry leaders (e.g., Google, Bosch, TCS). Different editions have highlighted themes such as mHealth,



This work is licensed under a Creative Commons Attribution 4.0 International License. *UbiComp Companion '25, Espoo, Finland*
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3750559>

smart textiles, music computing, and wireless sensing, reflecting the workshop's dynamic and interdisciplinary nature.

Apart from paper and poster presentation sessions, invited talks from academic and industry personnel, panels on general challenges in wearables research as well as focused topics such as mHealth, smart clothing and fabrics have been held so far. The prior versions of the workshop can be found below:

<https://www.eventcreate.com/e/bodysys24>

<https://bodysys.netlify.app/>

<https://bodysys-acm.com>

<https://sites.google.com/view/wearsys>

<http://www.wikicfp.com/cfp/servlet/event.showcfp?eventid=86780>

<https://sites.google.com/site/wearsys2018/home>

<http://mns.ucdenver.edu/wearsys2016/index.html>

<http://http://mobile.ece.msu.edu/acm-wearsys-2017/>

<https://sites.google.com/site/wearsys2015/home>

More information about the 2025 version of the workshop and its organizing committee can be found at: <https://www.eventcreate.com/e/bodysys25>.

3 Call for Papers

The 11th BodySys workshop comes at a critical juncture where body-centric computing-based research that includes sensing, communication, and computing-based systems has become prominent in the mobile computing space. Body centric computing has become more prominent than ever today with advancements in implantable computing, interest in through-body and intra-body communication and involving humans in the loop kind of systems. The use of smart sensing and diagnostic tools are becoming more common for monitoring animal pets and livestock (in smart agriculture).

BodySys will provide a venue for presenting current research and technology trends, and debating future research agendas of body-centric computing technology. It will provide a forum for discussing innovative and/or ideas that have potential for significant impact. We solicit papers of six or fewer pages that present preliminary research in broad areas of body-centric computing, with efforts inclusive of prototyping a system, experience in designing a novel technology, or survey of useful tools for designing inter-disciplinary systems and applications. We also encourage position papers that propose new directions for research or advocate disruptive design ideas and project applications. We also encourage submissions that can help bootstrap exploration of the body-centric computing space by the broader mobile systems community. The focus areas include, but not limited to,

- Human-in the loop systems
- Ubiquitous sensing systems
- Wearable in health, care, and well-being
- Large Language Models (LLMs) for humans
- Physical rehabilitation systems
- Human computer interaction (HCI) systems
- Sensing and Computing for Animals or Pets
- XR wearables, wearable imaging, projection and low-power displays
- Capacitive sensing and On-body communication

- Wearable health and wellness including activity monitoring, physiological sensing
- Novel energy management solutions (eg. swappable batteries, solar harvesting)
- Context-sensitive notification delivery
- Wearable biometrics for payment and authentication
- Haptics and cognitive prosthetics
- Body energy harnessing
- Brain-interfaces
- Electromyography (EMG) interfaces
- Wearable fashion (e.g. smart jackets, bodysuits, amulets), smart fabrics, smart patches
- AI on wearables: on-device ML/DL for wearable computing
- Novel combinations of 3D printing and Wearables (e.g., integration with custom 3D printed sensors)
- Wearable infrastructure and toolkits (hardware/software, custom/open source)
- Wearable in health, care, and well-being
- Wearable in ergonomics
- On-Body, Body-to-Body and Off-Body

4 Format and Program

The workshop will span half a day in two sessions, open to all participants. Session 1 will include a keynote and four paper presentations. Session 2 will include six paper presentations. We will also encourage presenters to set up their systems running at the beginning of the conference, so as to provide the presenters with a chance for more interactions with the audience.

BodySys 2025 will run as a hybrid event in the time zone of the conference's original venue (*i.e.*, Finland Standard Time). The conference will utilize Zoom to host the online participants. During the workshop event, the paper talks will be live. The presenter will join the live Zoom session to present and participate in the Q/A session.

5 Steering Committee Members

Jie Liu (Harbin Inst. of Technology, China)

Suman Banerjee (Univ. Wisconsin Madison, USA)

Mahadev Satyanarayanan (Carnegie Mellon Univ., USA)

Mi Zhang (Michigan State University, USA)

Ashwin Ashok (Georgia State University, USA)

6 Organizing Committee

6.1 General Co-Chairs

Qiang Yang is a Postdoctoral Research Associate in the Department of Computer Science and Technology at the University of Cambridge, UK. Prior to that, he received his Ph.D. from The Hong Kong Polytechnic University in 2023. His research interests lie in mobile health and ubiquitous computing, with a particular focus on designing novel models and systems that support well-being and healthier lifestyles through IoT and AI technologies. His work has been published in top-tier venues such as ACM MobiCom, MobiSys, SenSys, IMWUT, and CHI, and he was recognized as a Rising Star at ACM MobiSys 2025.

Thivya Kandappu is an Assistant Professor at the School of Computing and Information Systems at Singapore Management University (SMU). She received her PhD from University of New South Wales, Australia, in 2014. She has extensive experience in the mobile and ubiquitous computing and sensing research field – her specific research interests include designing, and building wearable devices for individualized monitoring of various human cognitive behaviors and behavioral analytics. Her works have been well recognized in top tier venues such as CSCW, UbiComp, MobiSys, and CHIIR. Kandappu was recognised as one of the top-200 most qualified Young Researchers to attend Heidelberg Laureate Forum 2018.

6.2 Technical Program Committee

Zhenlin An (University of Georgia, USA)
 Tao Chen (Samsung Research America, USA)
 Kaiyan Cui (Nanjing University of Posts and Telecommunications, China)

Ningning Hou (Macquarie University, Australia)
 Sijie Ji (California Institute of Technology, USA)
 Yang Liu (University of Cambridge, UK, **TPC Chair**)
 Yang Liu (Nokia Bell Lab, UK)
 Dong Ma (Singapore Management University, Singapore)
 Xiaomin Ouyang (HKUST, Hong Kong)
 Meera Radhakrishnan (University of Technology Sydney, Australia)
 Sougata Sen (BITS Pilani KK Birla Goa campus, India)
 Qijia Shao (HKUST, Hong Kong)
 Vigneshvaran Subbaraju (IHPC, A*STAR, Singapore)
 Yanwen Wang (Hunan University, China)
 Stephen Xia (Northwestern University, USA)
 Huanqi Yang (City University of Hong Kong, Hong Kong)
 Zhimeng Yin (City University of Hong Kong, Hong Kong)

6.3 Website/Publicity Chair

Yu Yvonne Wu (University of Cambridge, UK)